

Derivative

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1 Limits

To calculate the derivative, we look at the change over time h ,

$$\frac{f(x+h) - f(x)}{h} \quad h \neq 0$$

So for the function $f(x) = x^2$,

$$\begin{aligned} f'(x) &= \frac{(x+h)^2 - x^2}{h} \\ &= \frac{x^2 + 2xh + h^2 - x^2}{h} \\ &= \frac{2xh + h^2}{h} \\ &= \frac{h(2x + h)}{h} \\ &= 2x + h \end{aligned}$$

$$\lim_{h \rightarrow 0} f'(x) = 2x$$

For function $f(x) = \sqrt[3]{x}$,